

poojasri-v

/

python-programing-mechine-learning

<>

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

main

python-programing-mechine-learning / Time_series5.ipynb

Go to file

T

poojasri-v

Add files via upload

4062ca0 · 10 minutes ago

212 lines (212 loc) · 6.11 KB

Preview

Code

Blame

Raw

```
In [ ]: import pandas as pd
import numpy as np
from datetime import datetime
dates = [datetime(2020,1,5),
datetime(2020,1,10),
datetime(2020,1,15),
datetime(2020,1,20),
datetime(2020,1,25)]
ts = pd.Series(np.random.randn(5), index=dates)
print("\nWith this date object, create a series with datetime index:\n",ts)
```

With this date object, create a series with datetime index:

2020-01-05	0.616768
2020-01-10	-0.709351
2020-01-15	-0.250120
2020-01-20	1.473359
2020-01-25	0.769895

dtype: float64

```
In [ ]: res=ts.truncate(after="1/15/2020")
print("\nTruncate method to truncate datetime series \n", res)
```

Truncate method to truncate datetime series

2020-01-05	0.616768
2020-01-10	-0.709351
2020-01-15	-0.250120

dtype: float64

```
In [ ]: da=pd.date_range("10/1/2022", periods=10, freq="W-SUN")
print("\nDate range with day periods.\n",da)
```

Date range with day periods.

DatetimeIndex(['2022-10-02', '2022-10-09', '2022-10-16', '2022-10-23',
'2022-10-30', '2022-11-06', '2022-11-13', '2022-11-20',
'2022-11-27', '2022-12-04'],
dtype='datetime64[ns]', freq='W-SUN')

```
In [ ]: date=pd.DatetimeIndex(["1/1/2020","1/2/2020","1/2/2020","1/2/2020","1/3/2020"])
ts1=pd.Series(np.arange(5), index=date)
print("\nDuplicate datetime index:\n",ts1)
```

Duplicate datetime index:

2020-01-01	0
2020-01-02	1
2020-01-02	2
2020-01-02	3
2020-01-03	4

dtype: int64

```
In [ ]: group=ts1.groupby(level=0)
print("\nRepeated date counts:\n",group.count())
```

Repeated date counts:

2020-01-01	1
2020-01-02	3
2020-01-03	1

dtype: int64

```
In [ ]: long_df=pd.DataFrame(np.random.randn(10,4),index=da,columns=list("ABCD"))
print("\nMulti-column DataFrame with datetime index:\n",long_df)
```

Multi-column DataFrame with datetime index:

	A	B	C	D
2022-10-02	-0.131192	0.047824	-0.964743	-0.938759
2022-10-09	0.793063	-0.405336	-0.413008	0.659385
2022-10-16	1.024465	-1.410281	0.662419	-1.487978
2022-10-23	-0.656481	0.349419	-1.868689	1.093213
2022-10-30	1.014470	-1.939372	1.384576	2.386280
2022-11-06	0.144072	0.746535	1.282656	0.635330
2022-11-13	1.529731	-0.609798	1.407364	0.990335
2022-11-20	-0.812756	0.358335	-1.480371	1.158972
2022-11-27	-0.546305	1.541237	-0.516308	0.039178
2022-12-04	0.514115	-1.253606	1.488238	-0.519512